

Logs with zeros? Some problems and solutions*

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Abstract

When studying an outcome Y that is weakly-positive but can equal zero (e.g. earnings), researchers frequently estimate an average treatment effect (ATE) for a “log-like” transformation that behaves like $\log(Y)$ for large Y but is defined at zero (e.g. $\log(1+Y)$, $\operatorname{arcsinh}(Y)$). We argue that ATEs for log-like transformations should not be interpreted as approximating percentage effects, since unlike a percentage, they depend on the units of the outcome. In fact, we show that if the treatment affects the extensive margin, one can obtain a treatment effect of any magnitude simply by re-scaling the units of Y before taking the log-like transformation. This arbitrary unit-dependence arises because an individual-level percentage effect is not well-defined for individuals whose outcome changes from zero to non-zero when receiving treatment, and the units of the outcome implicitly determine how much weight the ATE for a log-like transformation places on the extensive margin. We further establish a trilemma: when the outcome can equal zero, there is no treatment effect parameter that is an average of individual-level treatment effects, unit-invariant, and point-identified. We discuss several alternative approaches that may be sensible in settings with an intensive and extensive margin, including (i) expressing the ATE in levels as a percentage (e.g. using Poisson regression), (ii) explicitly calibrating the value placed on the intensive and extensive margins, and (iii) estimating separate effects for the two margins (e.g. using Lee bounds). We illustrate these approaches in three empirical applications.

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1 Introduction

When the outcome of interest Y is strictly positive, researchers often estimate an average treatment effect (ATE) in logs of the form $E_P[\log(Y(1)) - \log(Y(0))]$, which has the appealing feature that its units approximate percentage changes in the outcome.¹ A practical challenge in many economic settings, however, is that the outcome may sometimes equal zero, and thus the ATE in logs is not well-defined. When this is the case, it is common for researchers to estimate ATEs for alternative transformations of the outcome such as $\log(1+Y)$ or $\operatorname{arcsinh}(Y) = \log(\sqrt{1+Y^2} + Y)$, which behave similarly to $\log(Y)$ for large values of Y but are well-defined at zero. The treatment effects for these alternative transformations are typically interpreted like the ATE in logs, i.e. as (approximate) average percentage effects. For example, among the 11 papers published in the *American Economic Review* since 2018 that interpret a treatment effect for $\operatorname{arcsinh}(Y)$, all but one interpret the result as a percentage effect or elasticity.²

The main point of this paper is that identified ATEs that are well-defined with zero-valued outcomes should not be interpreted as percentage effects, at least if one imposes the logical requirement that a percentage effect does not depend on the baseline units in which the outcome is measured (e.g. dollars, cents, or yuan).

Our first main result shows that if $m(y)$ is a function that behaves like $\log(y)$ for large values of y but is defined at zero, then the ATE for $m(Y)$ will be *arbitrarily sensitive* to the units of Y . Specifically, we consider continuous, increasing functions $m(\cdot)$ that approximate $\log(y)$ for large values of y in the sense that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. The common $\log(1+y)$ and $\operatorname{arcsinh}(y)$ transformations satisfy this property. We show that if the treatment affects the extensive margin (i.e. $P(Y(1) = 0) \neq P(Y(0) = 0)$), then one can obtain any magnitude for the ATE for $m(Y)$ by rescaling the outcome by some positive factor a . It is therefore inappropriate to interpret the ATE for $m(Y)$ as a percentage effect, since a percentage is inherently a unit-invariant quantity, while the ATE for $m(Y)$ depends arbitrarily on the units of Y .

The intuition for this result is that a “percentage” treatment effect is not well-defined for an individual for whom treatment increases their outcome from zero to a positive value. For example, in our application to Carranza et al. (2022) in Section 5, the treatment induces more people to have positive hours worked. The percentage change in hours is then not well-defined for individuals who would work positive hours under the treatment condition but zero hours under the control condition. Any average treatment effect that is well-defined with zero-valued outcomes must therefore implicitly assign a value for a change along the extensive margin. For logarithm-like transformations $m(\cdot)$, the importance of the extensive margin is determined implicitly by the units of Y . To see why this is the case, consider an individual who works positive hours only if they are treated, so that $Y(1) > 0$ and $Y(0) = 0$. Their treatment effect for the transformed outcome $m(Y)$ is $m(Y(1)) - m(0)$, which becomes larger if the units of Y are re-scaled by some $a > 1$, e.g. if we convert from weekly hours

¹That is, $\log(Y(1)/Y(0)) \approx \frac{Y(1)-Y(0)}{Y(0)}$ when $Y(1)/Y(0) \approx 1$.

²We found 17 papers overall using $\operatorname{arcsinh}(Y)$ as an outcome variable, of which 11 interpret the units; see Appendix Table 1.

effect. Our results thus cover the OLS estimator when it is consistent for the ATE for some (sub)-population P (e.g. in an RCT or under unconfoundedness). Given the prominence of OLS in applied work—and the fact that it is sometimes used for non-causal analyses—we now provide a direct result on the sensitivity to scaling of the estimand of an OLS regression of an outcome of the form $m(aY)$ on an arbitrary random variable X .

Specifically, suppose that $(X, Y) \sim Q$, for $Y \in [0, \infty)$ and $X \in \mathbb{R}^J$, where the first element of X is a constant. Consider the OLS estimand

$$\beta(a) = E_Q[XX']^{-1} E_Q[Xm(aY)],$$

i.e. the population coefficient from a regression of $m(aY)$ on X . We assume that $E_Q[XX']$ is full-rank so that $\beta(a)$ is well-defined. Letting $\beta_j(a) = e_j' \beta(a)$ be the j^{th} element of $\beta(a)$, we will show that $\beta_j(a)$ can be made to have arbitrary magnitude via the choice of a if $\gamma_j \neq 0$, where

$$\gamma = E_Q[XX']^{-1} E_Q[X \mathbb{1}[Y > 0]]$$

is the coefficient from a regression of $\mathbb{1}[Y > 0]$ on X .

Proposition 6. *Suppose that*

1. *The function m satisfies parts 1 and 2 of [Proposition 1](#).*
2. *(Finite expectations) $E_Q[\|X\|] < \infty$ and $E_Q[\|X \log(Y)\| \mid Y > 0] < \infty$.*

Then for every $j \in \{2, \dots, J\}$, $\beta_j(a)/\log(a) \rightarrow \gamma_j$ as $a \rightarrow \infty$. Moreover, if $\gamma_j \neq 0$ for some $j \in \{2, \dots, J\}$, then for every $\beta_j^ \in (0, \infty)$, there exists $a > 0$ such that $|\beta_j(a)| = \beta_j^*$. In particular $\beta_j(a)$ is continuous with $\beta_j(a) \rightarrow 0$ as $a \rightarrow 0$ and $|\beta_j(a)| \rightarrow \infty$ as $a \rightarrow \infty$.*

We note that [Proposition 6](#) implies that the OLS estimator for the j^{th} coefficient, $\hat{\beta}_j(a)$, will be arbitrarily sensitive to the choice of a when the corresponding extensive margin OLS estimator $\hat{\gamma}_j$, is non-zero. This follows immediately from setting Q to be the empirical distribution of $(Y_i, X_i)_{i=1}^N$ and applying [Proposition 6](#) (note that part 2 of the Proposition is trivially satisfied for the empirical distribution, since X and Y are both bounded over the empirical distribution).

OLS Standard Errors. We also show that as $a \rightarrow \infty$, the t -statistic for the OLS estimate $\hat{\beta}_j$ constructed using heteroskedasticity-robust standard errors converges to the t -statistic for $\hat{\gamma}_j$ (again using heteroskedasticity-robust standard errors). Formally, let

$$\hat{\Omega}_\beta(a) = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i X_i' \hat{\epsilon}_i(a)^2 \right) \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1}$$

denote the estimator of the heteroskedasticity-robust variance matrix for $\hat{\beta}(a)$, where $\hat{\epsilon}_i(a) = m(aY_i) - X_i' \hat{\beta}(a)$, and $\hat{\beta}(a)$ is the OLS estimate of $\beta(a)$. The t -statistic for $\hat{\beta}_j(a)$ is then $\hat{t}_{\beta_j}(a) =$

$\hat{\beta}_j(a)/\hat{\sigma}_{\beta_j}(a)$, where $\hat{\sigma}_{\beta_j}(a) = \sqrt{e_j' \hat{\Omega}_{\beta}(a) e_j} / \sqrt{N}$. Analogously, let

$$\hat{\Omega}_{\gamma} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i X_i' \hat{u}_i^2 \right) \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1}$$

be the heteroskedasticity-robust variance estimator for $\hat{\gamma}$, the OLS estimate of γ , where $u_i = \mathbb{1}[Y_i > 0] - X_i' \hat{\gamma}$. The t -statistic for $\hat{\gamma}_j$ is then $\hat{t}_{\gamma_j} = \hat{\gamma}_j / \hat{\sigma}_{\gamma_j}$, where $\hat{\sigma}_{\gamma_j} = \sqrt{e_j' \hat{\Omega}_{\gamma} e_j} / \sqrt{N}$.

Proposition 7. *Suppose that $(\frac{1}{N} \sum_i X_i X_i')$ is full-rank and that $\hat{\sigma}_{\gamma_j} > 0$. If the function m satisfies parts 1 and 2 of [Proposition 1](#) and $\hat{\gamma}_j > 0$, then $\hat{t}_{\beta_j}(a) \rightarrow \hat{t}_{\gamma_j}$ as $a \rightarrow \infty$.*

It follows that when the units of Y are made large, the t -statistic for a treatment effect estimate for $m(Y)$ estimated using OLS will converge to the t -statistic for the OLS estimate of the extensive margin. [Appendix Figure 1](#) shows that, indeed, the t -statistics for estimates using $\text{arcsinh}(Y)$ in the *AER* tend to be close to the t -statistics for the extensive margin, and tend to become even closer after rescaling the units by a factor of 100.

Proof of Proposition 6. Fix $j \in \{2, \dots, J\}$. Note that $\beta(0) = E_Q[XX']^{-1} E[Xm(0)]$, is the coefficient from a regression of a constant outcome $m(0)$ on X , and thus $\beta_1(0) = m(0)$ while $\beta_k(0) = 0$ for $k \geq 2$. Thus $\beta_j(0) = 0$. To complete the proof, we will first show that $\beta_j(a_n) = \gamma_j \log(a_n) + o(\log(a_n))$. Hence, if $\gamma_j > 0$, then $|\beta_j(a)| \rightarrow \infty$ as $a \rightarrow \infty$. We will then establish that $\beta_j(a)$ is continuous for $a \in [0, \infty)$. The fact that one can obtain any positive value for $|\beta_j(a)|$ then follows from the intermediate value theorem.

For ease of notation, let $\nu' = e_j' E_Q[XX']^{-1}$, so that $\beta_j(a) = E_Q[\nu' X m(aY)]$.

We first show that $\beta_j(a_n) = \gamma_j \log(a_n) + o(\log(a_n))$. Consider a sequence $a_n \rightarrow \infty$, and assume without loss of generality that $a_n > e$. Let $f_n(x, y) = \nu' x \cdot m(a_n y) / \log(a_n)$. Observe that $f_n(x, y) \rightarrow \nu' x \cdot \mathbb{1}[y > 0]$ pointwise, since $f_n(x, 0) = \nu' x \cdot m(0) / \log(a_n) \rightarrow 0$, while for $y > 0$,

$$f_n(x, y) = \nu' x \cdot \frac{m(a_n y)}{\log(a_n)} = \nu' x \cdot \frac{m(a_n y)}{\log(a_n y)} \frac{\log(a_n) + \log(y)}{\log(a_n)} \rightarrow \nu' x,$$

where we use the fact that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$. We showed in the proof to [Proposition 4](#) that

$$\frac{|m(a_n y)|}{\log(a_n)} \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot (1 + |\log(y)|),$$

which implies that

$$|f_n(x, y)| \leq |\nu' x \cdot (\kappa + 2 \cdot \mathbb{1}[y > 0] \cdot (1 + |\log(y)|))| =: \bar{f}(x, y).$$

Moreover, part 2 of the proposition implies that $\bar{f}(X, Y)$ is integrable. From the dominated convergence theorem, it follows that

$$\frac{\beta_j(a_n)}{\log(a_n)} = E_Q[f_n(X, Y)] \rightarrow E_Q[\nu' X \mathbb{1}[Y > 0]] = \gamma_j.$$

incompatible with $P(Y(1) = 0) > 0$ if $\beta_1 > 0$. By similar logic, the model is also incompatible with $P(Y(0) = 0) > 0$ if $\beta_1 < 0$. In settings where there is point-mass at zero, the model that [Bellemare and Wichman \(2020\)](#) show gives β_1 an interpretation as a semi-elasticity will therefore typically be rejected by the data. It is also worth noting that even if there are no zeros in the data, the model in (C.2) will generally be sensitive to units, in the sense that if (C.2) holds for Y measured in dollars, it will generally not hold when Y is measured in cents. The validity of the interpretation of β_1 as an elasticity thus depends on having chosen the “correct” scaling of the outcome such that (C.2) holds.

Similar issues apply if we consider alternative transformations on the left-hand side of (C.1). For example, [Thakral and Tô \(2023\)](#) consider versions of (C.1) that replaces $\text{arcsinh}(Y)$ with the power function Y^k . They then consider the implied “semi-elasticities” of the form $\eta(y_0) = \beta_1 / (k y_0^k)$. The parameter $\eta(y_0)$ has the interpretation as a structural semi-elasticity when d has a constant effect on Y^k . Specifically, if D is continuous and the structural equation

$$Y(d, U)^k = \beta_0 + d\beta_1 + U, \quad (\text{C.3})$$

holds, then $\eta(y_0) = (\frac{\partial}{\partial d} Y(d, U)) / Y(d, U)$ evaluated at $Y(d, U) = y_0$, so $\eta(y_0)$ corresponds to the semi-elasticity of $Y(d, U)$ with respect to d . However, as with (C.2), (C.3) is generally incompatible with settings in which $P(Y(d, U) = 0)$ for multiple values of d . For example, if $\beta_1 > 0$, then $Y(0, U) \geq 0$ implies that $Y(1, U) > 0$. Equation (C.3), which gives a causal interpretation to $\eta(\beta_0)$ as a semi-elasticity, will thus generally be incompatible with settings in which some units have $Y = 0$ under multiple treatment statuses.

C.2 [Cohn et al. \(2022\)](#)

[Cohn et al. \(2022\)](#) consider structural equations of the form

$$Y = \exp(\alpha + D\beta)U. \quad (\text{C.4})$$

When $E[U \mid D] = 1$, they show that Poisson regression is consistent for β , whereas regressions of $\log(1 + Y)$ or $\log(Y)$ on D may be inconsistent for β .⁵² Although [Cohn et al. \(2022\)](#) do not consider a potential outcomes interpretation of β , we can give β a causal interpretation if we impose that the potential outcomes take the form

$$Y(d, U) = \exp(\alpha + d\beta)U(d), \quad (\text{C.5})$$

where $E[U(d)] = 1$. Under (C.5), it follows that $\exp(\beta) = E[Y(1)]/E[Y(0)]$, i.e. the parameter

⁵²We thank Kirill Borusyak for an insightful discussion on this topic. Relatedly, in an influential paper, [Santos Silva and Tenreiro \(2006\)](#) consider the structural equations model $Y_i = \exp(X_i'\beta)U_i$ where $E[U_i \mid X_i] = 1$, and show that Poisson regression consistently estimates β while a regression using log on the left-hand side does not, although they do not provide any formal results on log-like transformations.